

Dark Energy as the Vacuum Energy of the MOND Field: a Consistent Model with a Turn-Off Near 10^{-8} m s^{-2}

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Abstract

The MOND acceleration scale and the cosmological constant obey $\Lambda \sim a_0^2/c^4$ to within an order-one factor. We propose that this is not a coincidence: the observed dark energy is the vacuum (zero-field) energy of the field that produces MOND. In the field theories of MOND that have a Lagrangian (QUMOND, and the nonrelativistic limit of bimetric MOND), the zero-field energy is fixed by the interpolating function, $\Lambda/a_0^2 = \frac{1}{2} \int_0^\infty (\nu - 1) d(y^2)$ with $y = g_N/a_0$, identically in both bimetric classes, and it is positive, i.e. de Sitter. For the quadrature kernel $\nu = \sqrt{1 + 1/y}$ the integral diverges, and the same slow tail violates the planetary ephemerides; a strong-field turn-off $\nu - 1 \rightarrow (\sqrt{1 + 1/y} - 1)/[1 + (y/y_t)^2]$ cures both and gives $\Lambda/a_0^2 \simeq (\pi/4) y_t$. With measured $a_0 = 1.097 \times 10^{-10} \text{ m s}^{-2}$, the observed Λ then fixes $y_t = 94$ (71–117 at 2σ), a turn-off at $g_N \approx 1.0 \times 10^{-8} \text{ m s}^{-2}$. This is allowed by SPARC rotation curves and by every inner planet. The model predicts (i) $w = -1$ together with a time-constant a_0 , or $a_0 \propto \sqrt{\rho_{\text{DE}}}$ if dark energy evolves; (ii) the MOND boost turning off near 10^{-8} m s^{-2} , a signature of $\lesssim 0.002 \text{ dex}$ that no galaxy-scale probe can reach; (iii) no quantum zero-point origin for Λ . It does not derive the coefficient: a Lean proof shows the relation $a_0 = \frac{1}{2} c \sqrt{G \rho_\Lambda}$ is independent of the other ingredients, and the turn-off scale is a new constant. The cold mass of cosmology ($\Omega_c h^2 \approx 0.12$) is still required.

1 The proposal

Two scales of very different origin are numerically close: MOND's $a_0 \approx 1.1 \times 10^{-10} \text{ m s}^{-2}$, fitted to galaxies [LMS16, D23], and $c^2 \sqrt{\Lambda}$ from cosmology. Milgrom has emphasised the coincidence [M20]. Our proposal is that dark energy *is* the vacuum energy of the MOND field: the energy density left in its Lagrangian when the field vanishes.

2 The vacuum energy of a MOND field theory

In QUMOND the Lagrangian density is $-\frac{1}{8\pi G} [2\nabla\Phi \cdot \nabla\Phi_N - a_0^2 Q(|\nabla\Phi_N|^2/a_0^2)]$ with $Q'(z) = \nu(\sqrt{z})$. Requiring no vacuum constant in strong fields ($Q(z) - z \rightarrow 0$), the zero-field value is $Q(0) = -\int_0^\infty (\nu - 1) dz$. The same structure appears in Milgrom's bimetric theory [M09]: its vacuum term (his eq. 24) at equal metrics is $\Lambda = -\frac{1}{2}(1 + f'(1)) a_0^2 M(0)$, with $M' = \nu - 1$ in the nonrelativistic limit (his eq. 3). Its sign is automatically de Sitter. We derived the map for the exchange-symmetric class ($\alpha = \beta$), $M'(z) = (\nu - 1)/(2\nu - 1)$ at $z = [y(2\nu - 1)]^2$. The two classes' vacuum integrals differ by the boundary term $2y^2(\nu - 1)^2$, which vanishes. So, with $f'(1) = 0$ (fixed by exchange symmetry),

$$\boxed{\frac{\Lambda c^4}{a_0^2} = \frac{1}{2} \int_0^\infty (\nu - 1) d(y^2)} \quad (1)$$

is a property of the force law alone. The algebra (vacuum condition, exchange symmetry pinning $f'(1) = 0$, the class identity) is certified in Lean with standard axioms.

3 The tail, and why it must turn off

For $\nu = \sqrt{1 + 1/y}$ (the quadrature law $g^2 = g_N^2 + a_0 g_N$, Milgrom 1999 [M99]) one has $\nu - 1 \simeq 1/(2y)$, so the integral grows linearly with any cutoff. Cut at the Planck acceleration it overshoots by $\sim 10^{59}$. The same tail gives a constant $a_0/2$ sunward anomaly, 1279 times the Earth–Mars ephemeris bound [SJ06]. One turn-off cures both:

$$\nu(y) = 1 + \frac{\sqrt{1 + 1/y} - 1}{1 + (y/y_t)^2}, \quad \frac{\Lambda c^4}{a_0^2} \simeq \frac{\pi}{4} y_t. \quad (2)$$

The law is unchanged below y_t . SPARC cannot distinguish it from the exact law for $y_t \geq 100$ ($|\Delta\chi^2| \leq 1.06$), and a direct fit bounds y_t only from below (≥ 2 –20, depending on the mass-to-light treatment). All inner planets pass for $y_t \leq 7.7 \times 10^5$ (Mars binds).

4 What the observed Λ fixes

With measured a_0 the observed ratio is $\Lambda c^4/a_0^2 = 73.19$ (ensemble of nine SPARC analyses, $a_0 = 1.097 \times 10^{-10}$, $\pm 12\%$), i.e. 55.3–91.1 at 2σ . Solving the vacuum condition (Fig. 1) gives

$$y_t = 94 \text{ (71–117 at } 2\sigma\text{)}, \quad g_t = y_t a_0 \approx 1.0 \times 10^{-8} \text{ m s}^{-2}. \quad (3)$$

On the footing that defines $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$ the same condition gives $y_t = 128.9$. Either way the turn-off sits just above the highest accelerations in SPARC ($y \lesssim 110$) and far below the planets ($y \sim 10^7$ – 10^8).

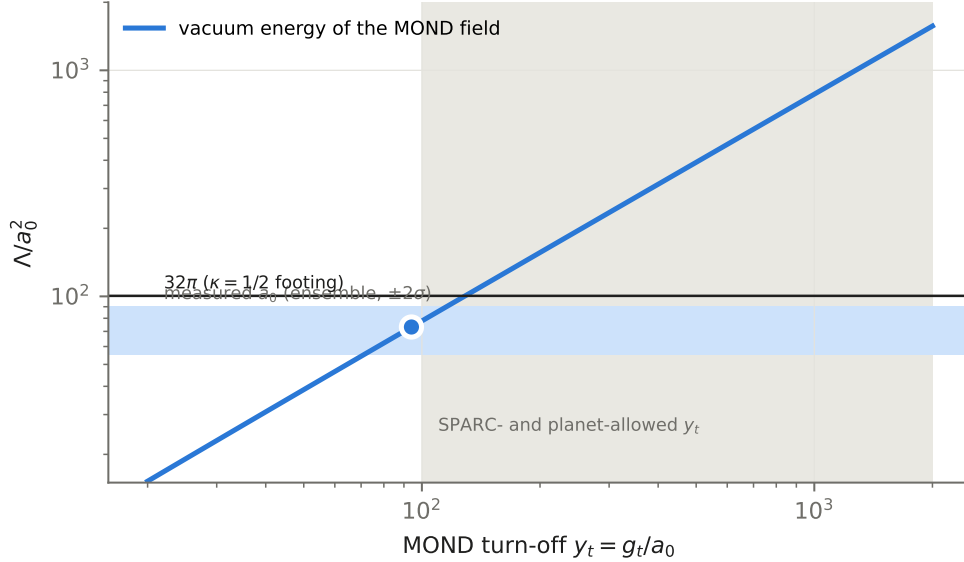


Figure 1: The MOND field’s vacuum energy $\Lambda c^4/a_0^2$ against its turn-off y_t (blue). Horizontal band: the value required by the observed Λ with measured a_0 ($\pm 2\sigma$); the dot marks the central solution $y_t = 94$. Black line: the $\kappa = \frac{1}{2}$ footing. Shaded: turn-offs allowed by SPARC and the planets.

5 Predictions

- Equation of state.** If the kernel is fixed, $\rho_{\text{DE}} \propto a_0^2$ at all times. A constant a_0 means $w = -1$ exactly. Conversely, if dark energy evolves (as DESI DR2 hints), $a_0(z) \propto \sqrt{\rho_{\text{DE}}(z)}$: about $0.8\times$ today’s value at $z = 2.5$ on DESI’s $w_0 w_a$ fits. The rival reading $a_0 \propto H(z)$ gives $3.7\times$. The two are separable with ~ 0.1 dex at $z \approx 2.5$.
- A MOND turn-off near 10^{-8} m s^{-2} , which is structurally out of reach.** Above g_t the boost falls as y^{-3} . Near y_t the boost itself is only $\nu - 1 \simeq 1/(2y_t) \approx 0.5\%$, so no galaxy-scale probe can see more than ~ 0.002 dex of the turn-off. For the 187 ATLAS3D early-type galaxies (JAM quality ≥ 1 , $y \leq 38$ at $r_{1/2}$) the predicted dynamical masses move by $\sim 10^{-4}$ dex, 60 times below the sample’s statistical reach and about 900 times below the 0.10 dex stellar-population floor; SLACS lenses ($y \approx 5$ –20) move by $\leq 5 \times 10^{-4}$ dex (programme lane p38). The planets show only that some turn-off exists ($y_t \leq 7.7 \times 10^5$). So g_t cannot be measured independently, and Prediction 1 is the model’s testable content.
- No zero-point Λ .** The cosmological constant is the MOND field’s energy, of order a_0^2/G , not a quantum zero-point sum, so the 10^{122} mismatch does not arise in this reading. Any separate matter vacuum energy must be cancelled or small.

4. **Galaxies need no dark-matter particle** (the MOND field does that work). Cosmology still needs the cold mass.

6 What is not derived

- **The coefficient.** A Lean proof shows that $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$ (equivalently $\Lambda A_S = 32\pi^2$ with A_S the area of the Schwarzschild horizon of surface gravity a_0) is *independent* of the ingredients above. Models exist with and without it, and every coefficient $\kappa > 0$ is realised. Here it is replaced by the turn-off y_t , a new constant.
- **The turn-off's shape.** Steeper turn-offs ($k = 3, 4$) change y_t for the same Λ (167.5 and 182.4 on the $\kappa = \frac{1}{2}$ footing).
- **The identification itself** (dark energy = MOND vacuum energy) is a hypothesis, not a consequence of the action.
- **The kernel's transition shape.** On SPARC the RAR kernel fits better than $\sqrt{1 + 1/y}$, but the preference is degenerate with the stellar mass-to-light ratio (it falls from $\Delta\chi^2 = 141$ to 12 as Υ_{disk} goes from 0.5 to 0.7).

Reproducibility. Scripts `sonnet55_push/puzzle_32pi/p21_bimond_vacuum.py`, `p25_symmetric_bimond_map.py`, `p35_kernel_tail_fix.py`, `p36_sparc_turnoff_fit.py`, `p37b_shape_vs_upsilon.py`; Lean `fable_independent_2026/lean_2026/PUZZLE_32pi_chain_2026_10_05.lean`, `PUZZLE_32pi_postulate_status_2026_10_05.lean`; figure `make_paper42_figures.py`; audit `PAPER42_audit.py`.

References. [D23] Desmond, arXiv:2303.11314 (2023). [LMS16] Lelli, McGaugh & Schombert, AJ 152, 157 (2016). [M99] Milgrom, Phys. Lett. A 253, 273 (1999), astro-ph/9805346. [M09] Milgrom, Phys. Rev. D 80, 123536 (2009), arXiv:0912.0790. [M20] Milgrom, arXiv:2001.09729 (2020). [SJ06] Sereno & Jetzer, MNRAS (2006).